

Week 3 Report

Training Three Classification Models

Logistic Regression, Decision Tree and Random Forest

Abstract

This report presents the model training work completed in Week 3 for a binary churn classification problem. Three classification models—Logistic Regression, Decision Tree, and Random Forest—were trained after applying the required preprocessing steps. The data was divided into training and testing sets using an 80/20 stratified split so that the class distribution remained similar in both sets. Since the dataset contained fewer churn cases than non-churn cases, class weighting was also tested. The initial results showed that accuracy alone did not give a clear picture of model performance. After balancing the classes, Logistic Regression reached a recall of 0.6817 and an F1-score of 0.5089, while Decision Tree reached the highest recall of 0.7183. Random Forest showed very little improvement from class weighting. Based on the Week 3 results, Logistic Regression is the current leading model, although final model selection will be completed after the evaluation planned for Week 4.

Keywords

Machine Learning, Classification, Churn Prediction, Logistic Regression, Decision Tree, Random Forest, Class Imbalance

Introduction

The aim of this project is to use machine learning to identify members who are likely to churn. Being able to identify these members can help an organization focus its retention efforts on people who may be at risk of leaving.

The Week 3 stage focuses on training three classification models and comparing their initial performance. An important issue in the data is class imbalance because churners represent a smaller part of the dataset. For this reason, the analysis looks beyond accuracy and gives particular attention to precision, recall, and F1-score.

Dataset and Target Variable

The target variable used for this classification task is Churn. It was converted from Yes/No values into a binary format where Yes is represented by 1 and No by 0. The reported churn rate is 26.5%, which means the two classes are not evenly distributed.

An 80/20 train-test split was used with stratification. This keeps the proportion of churn and non-churn cases similar in the training and testing data and provides a fairer basis for model comparison.

Data Preparation

Before training the models, the data was prepared according to the type of feature. Missing numeric values were handled using median imputation, followed by standard scaling. Categorical missing values were handled using the mode, and categorical variables were converted into numeric form using one-hot encoding. The encoding step used `handle_unknown='ignore'` so that new categories would not cause problems when the model is used later.

The preprocessing steps were placed inside a separate Pipeline for each model. This is important because the preprocessing parameters are learned from the training data rather than from the complete dataset, helping to prevent data leakage.

Models Used

Three classification algorithms were selected for comparison. Logistic Regression was used as a straightforward classification model and provides a useful baseline. Decision Tree was included because it can learn non-linear decision rules. Random Forest was used as an ensemble approach that combines multiple decision trees and can capture more complex patterns.

Methodology

Step	Approach
Target encoding	Churn: Yes → 1, No → 0
Train-test split	80/20 stratified split
Numeric features	Median imputation and StandardScaler
Categorical features	Mode imputation and One-Hot Encoding
Unknown categories	<code>handle_unknown='ignore'</code>
Class imbalance	<code>class_weight='balanced'</code>
Leakage prevention	Preprocessing included inside each Pipeline

Baseline Model Results

The first experiment was carried out without class balancing. The purpose was to see how the models performed under their default settings and to understand whether accuracy was giving a misleading impression of performance.

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.7615	0.5704	0.2169	0.3143
Decision Tree	0.7601	0.5419	0.3099	0.3943
Random Forest	0.7509	0.5130	0.2225	0.3104

All three models produced accuracy close to 75%, but their recall was much lower. They identified only around 22–31% of the actual churners. A simple classifier that predicted nobody as a churner

would achieve about 73.5% accuracy. This shows that accuracy by itself is not enough for this problem, because failing to identify a real churner can mean missing an opportunity to take retention action.

Results After Class Balancing

To give greater importance to the minority churn class, the three models were retrained using `class_weight='balanced'`. The results changed considerably for Logistic Regression and Decision Tree.

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.6686	0.4060	0.6817	0.5089
Decision Tree	0.6352	0.3812	0.7183	0.4980
Random Forest	0.7601	0.5603	0.2225	0.3185

Results Comparison and Discussion

The biggest change can be seen in recall. Logistic Regression increased from 0.2169 to 0.6817, while Decision Tree increased from 0.3099 to 0.7183. Their F1-scores also improved. At the same time, their accuracy decreased because the models became more willing to classify observations as churners.

Random Forest behaved differently. Its recall remained at 0.2225 after class weighting and its F1-score changed only slightly from 0.3104 to 0.3185. The Week 3 analysis also tested a lower prediction threshold for Random Forest. Using a threshold of 0.40 instead of 0.50 was reported to increase recall from 0.22 to 0.38 and F1-score to 0.43. A more complete threshold comparison is planned for Week 4.

Based on the current results, Logistic Regression is the leading model for Week 3. It has an F1-score of 0.5089 and recall of 0.6817. Decision Tree has slightly higher recall at 0.7183, but its precision and F1-score are lower. Therefore, Logistic Regression currently provides the better overall balance between precision and recall.

Challenges and How They Were Handled

The main challenge was the imbalance between churn and non-churn cases. If this imbalance is ignored, a model can obtain reasonable accuracy while missing a large number of churners. To address this, `class_weight='balanced'` was applied to all three models. Another concern was data leakage, so the imputation, encoding, scaling, and model were kept together in a Pipeline.

Deliverables

Three trained model pipelines were produced and saved for later use: LogisticRegression.pkl, DecisionTree.pkl, and RandomForest.pkl. Each saved file contains the preprocessing steps together with the model, which means the same data preparation can be applied when making future predictions. The models were also reloaded successfully and reproduced their predictions.

Limitations

The results in this report are from the Week 3 training stage and should not be treated as the final model evaluation. The current analysis does not yet include the complete Week 4 evaluation, such as detailed confusion matrices, ROC-AUC, and a full precision-recall threshold analysis. The current model ranking may therefore change after that evaluation.

Future Work

The next stage is to evaluate the three models in more detail using precision, recall, F1-score, confusion matrices, and ROC-AUC. Different prediction thresholds should also be compared, especially for Random Forest, to understand the trade-off between finding more churners and generating more false positives. After the evaluation, the final model can be selected and used in the prediction interface planned for the next stage.

Conclusion

In Week 3, three classification models were successfully trained and compared for the churn prediction task. The baseline experiment showed that accuracy alone can hide poor detection of churners. Class balancing substantially improved recall for Logistic Regression and Decision Tree, while Random Forest showed little change. Logistic Regression currently gives the strongest overall F1-score with good recall and is therefore the leading model at this stage. The final decision will be made after the detailed evaluation in Week 4.